Sandeep Roy

Jorhat Engineering College, Assam, India
Phone: +91-9395472165 Email: sandeeproy4984@gmail.com LinkedIn: linkedin.com/in/sandeeproyy

Career Objective

Undergraduate B.Tech student in Mechanical Engineering aiming to contribute to the advancement of intelligent robotics by developing innovative autonomous systems that integrate mechanical engineering, Embodied AI, computer vision, and machine learning. Experienced in CAD/CAE modelling, 3D printing, ROS-based robotics, embedded systems, SLAM-based autonomous navigation, and deep learning frameworks including TensorFlow and PyTorch. My ambition is to design, prototype, and deploy smart robotic platforms capable of operating effectively in complex real-world environments, while working at the intersection of robotics, artificial intelligence, and product innovation to create technologies with meaningful impact on industry and society.

Education

Degree	Institute	CGPA / %	Year
B.Tech in Mechanical Engineering	Jorhat Engineering College, Assam	7.21 / 10 (upto 5^{th} sem)	2027
AHSEC (Higher Secondary)	Lokanayak Omeo Kumar Das College, Dhekiajuli	80.80%	2023
SEBA (HSLC)	Missa Model High School, Nagaon	87.16%	2021

Experience

Research & Development Intern March 2024 – Present
Agronexa Solutions *On-Site*

- Collaborated with the CTO to architect and prototype a Smart IoT Device for Precision Farming and Irrigation, integrating sensor fusion (soil-moisture, temperature, humidity) with actuator control over MQTT.
- Conducted feasibility and system-architecture research for an Autonomous Agricultural Drone capable of variable-rate spraying, crop-health imaging (NDVI), and GPS-guided waypoint navigation.
- Secured top-10 ranking at Assam Startup Cohort 6.0 Idea Pitching competition with an AgriTech pitch covering market analysis, MVP design, and go-to-market strategy.
- Performed domain research identifying systemic inefficiencies in the Indian Tea Industry, mapping pain-points to potential IoT and ML-driven solutions.

Research & Development Intern June – July 2025
IIEST Shibpur *Online*

- Designed a Total Ossicular Replacement Prosthesis (TORP) for middle-ear reconstruction, selecting biocompatible materials (titanium alloy) based on biocompatibility, acoustic impedance, and fatigue-life constraints.
- Performed structural FEA (von Mises stress, modal analysis) in Autodesk Fusion 360 to validate mechanical integrity under physiological loading conditions.

Winter Research Intern December 2024 – January 2025
Technology Innovation Hub, IIT Guwahati *On-Site*

- Conducted experimental characterization of Ionic Polymer-Metal Composite (IPMC) actuators under low-voltage AC excitation across WATER, AIR, and 1% ACETONE environments.
- Developed an ESP32-CAM-based image-acquisition setup for real-time deformation monitoring of cantilever-configured IPMC actuators at 200 ms frame intervals.
- Implemented a computer-vision pipeline using contour analysis, skeletonization, tip tracking, and vector-angle estimation to extract bending deformation from image sequences.
- Generated a structured dataset of 840 observations across 42 experimental runs and trained a TensorFlow regression model achieving $R^2 = 0.959$ for deformation prediction.

Projects

GPS-Denied Autonomous Surveillance Drone | *IRoC-U 2026 Submission* October 2023 – Present

- Architecting a fully autonomous UAV that operates without GPS by fusing LiDAR point clouds, stereo-camera depth maps, and IMU data through an Extended Kalman Filter (EKF) for state estimation.
- Implemented Visual-Inertial Odometry (VIO) pipeline using ORB-SLAM3 running on an onboard NVIDIA Jetson, achieving real-time 6-DOF pose estimation at 30 Hz.
- Integrated ROS 2 middleware for modular inter-process communication between perception, planning (A*/RRT*), and control nodes; deployed PX4 autopilot firmware on flight controller.
- Used SLAM (Simultaneous Localisation and Mapping) with Computer Vision for autonomous obstacle detection, avoidance, and target tracking in GPS-denied indoor and semi-outdoor environments.

6-DOF Self-Designed & 3D-Printed Robotic Arm 2024 – Present

- Performed kinematic analysis (forward and inverse kinematics via Denavit-Hartenberg parameterisation) and dynamic modelling (Newton-Euler recursive formulation) to size servomotors and structural members.
- Designed all linkages, joints, and mounting brackets in SolidWorks with DFM constraints targeting FDM 3D printing in PETG; post-processed with ANSYS Static Structural to verify safety factors under rated load.
- Implemented joint-space trajectory planning and PID-based servo control via an Arduino Mega; exposed a ROS action-server interface for high-level task commanding.

IPMC Material Deformation Prediction Model January 2026 – May 2026

- Conducted experimental characterisation of Ionic Polymer-Metal Composite (IPMC) actuators under AC excitation (2.1 V – 4.9 V) in WATER, AIR, and 1% ACETONE environments; acquired 840 deformation observations across 42 experimental runs using an ESP32-CAM-based image acquisition system.
- Built a computer-vision pipeline (Python + OpenCV) for automated bending-angle extraction from sequential image frames via contour detection and affine-transform fitting.
- Trained a deep neural network (TensorFlow/Keras) regressor on the structured dataset; achieved $R^2 = 0.959$ between measured and predicted deformation responses, validated via k-fold cross-validation and residual analysis.
- Performed phase-space analysis, stimulus/relaxation/back-relaxation characterisation, and voltage-dependent actuation profiling using NumPy, Pandas, and Matplotlib on Google Colab.

Solar-Powered Attendance Management & Worker Tracking System | GitHub 2024

- Designed a solar-harvested, off-grid edge device integrating RFID/NFC-based attendance logging with a BLE wearable for real-time worker location tracking on factory/field sites.
- Implemented a lightweight ML classifier on the embedded microcontroller for anomaly detection in worker movement patterns; synced data to a cloud dashboard via MQTT over 4G/LTE fallback.

Skills

CAD / CAE	SolidWorks, Autodesk Fusion 360, ANSYS Workbench (Static Structural, Modal, CFD)
Programming	Python, C, C++, MATLAB, Bash, Git / GitHub
Machine Learning	TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, Pandas, Matplotlib, Seaborn
Computer Vision	OpenCV, MediaPipe, ORB-SLAM3, visual-inertial SLAM, LiDAR SLAM
Robotics	ROS / ROS 2, PX4 Autopilot, ArduPilot, Arduino, NVIDIA Jetson
Manufacturing	FDM 3D Printing (PLA, PETG, ABS), Laser Cutting, Rapid Prototyping
Tools	Docker, Google Colab, VS Code, Linux

Workshops and Certifications

National Embetronix Challenge — 1st Position 2024
Kshitij, IIT Kharagpur

Col. Guru Prasad Das Hackathon (JEC Alumni & Oil India Ltd.) — Winner 2025
Jorhat Engineering College

National Amphibian Bot Challenge — Winner 2024
IIT Kharagpur

National Line Follower Bot Challenge — Winner 2024
IIT Kharagpur

Specialization: Aerial Robotics 2025
University of Pennsylvania (Coursera)

Supervised Machine Learning: Regression and Classification 2025
Stanford Online (Coursera)

AI with Jetson NANO 2024
NVIDIA Deep Learning Institute

Generative AI and LLMs on Google Cloud 2023
Google Cloud Skills Boost

Python for Data Science, AI and Development 2024
IBM (Coursera)

Certificate of Completion — Winter Internship 2024
IIT Guwahati Technology Innovation Hub

Certificate of Completion — Summer Internship 2025
IIEST Shibpur

AI/ML and Robotics Workshop 2024
IIT Guwahati

3-Day Aeromodelling Workshop 2024
Jorhat Engineering College

Positions of Responsibility

Training and Placement Coordinator 2026 – 27
Jorhat Engineering College

Club Head 2025-26
Astronomy Society, Jorhat Engineering College

Research & Development Head 2025-present
Robotics Society, Jorhat Engineering College

Research & Development Head 2025-present
Electronics Society, Jorhat Engineering College

Executive Member 2023-present
Team Arcelor (Robotics Team), Jorhat Engineering College

Student Member 2025-present
Bureau of Indian Standards (BIS)